

Planning Scanner Capacity and Staffing in Digital Pathology: A Trace-Driven Simulation Framework

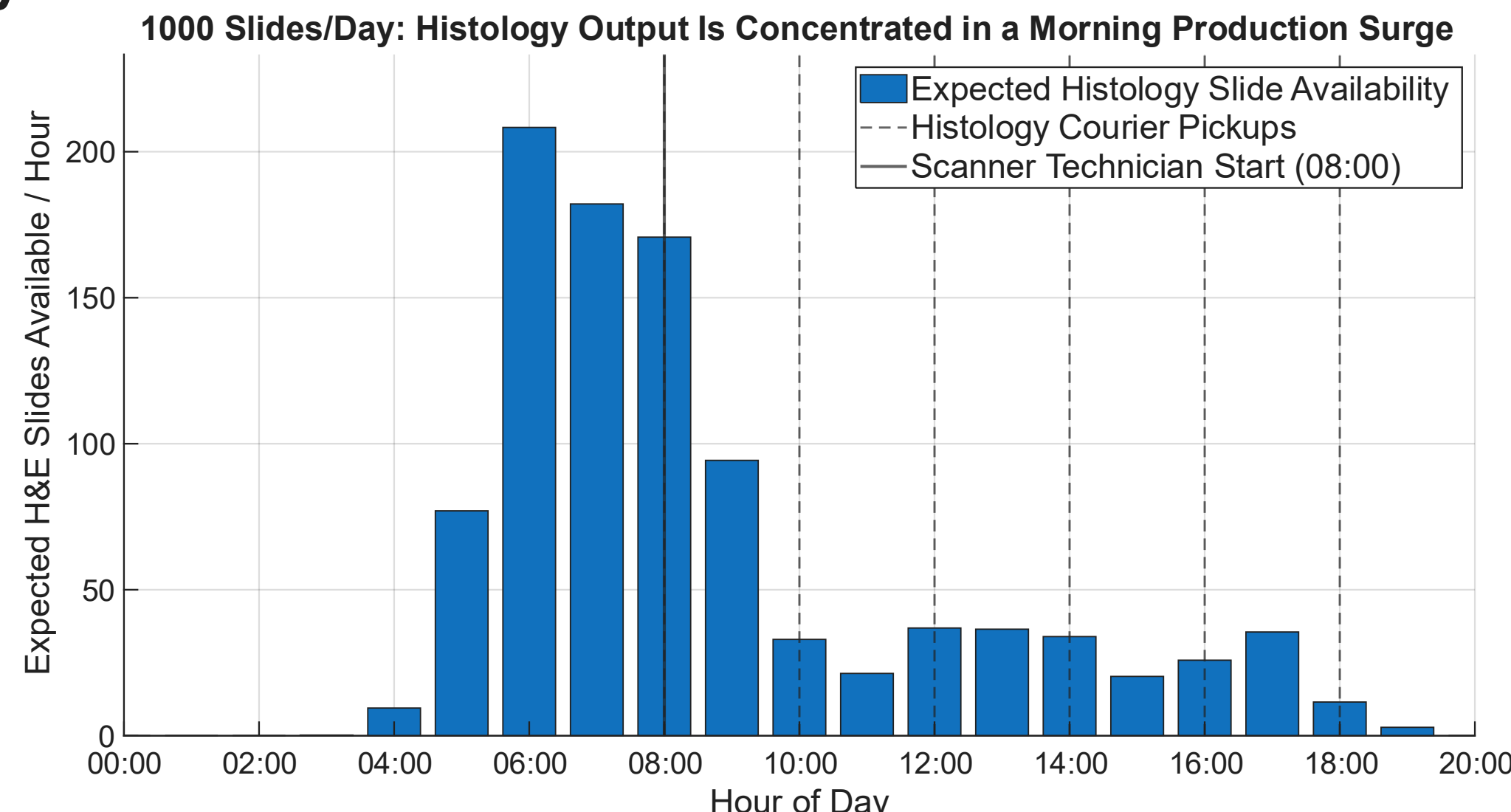
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OBJECTIVES

To quantify how time-dependent histology output, scanner fleet size, and scanner technician availability interact to determine same-day digital slide availability.

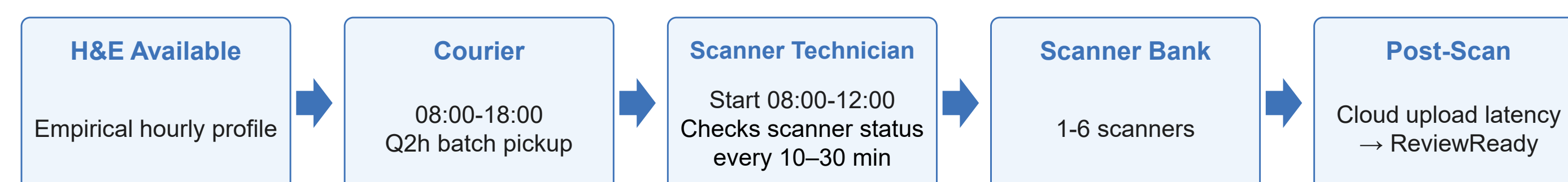


ABSTRACT / INTRODUCTION

The transition to digital pathology introduces complex operational challenges in **slide scanning, staffing, and turnaround time**. While whole slide imaging (WSI) adoption is accelerating, few quantitative tools exist to model workflow performance or **prospectively identify bottlenecks**. We developed a **trace-driven, discrete-event simulation (DES)** framework to model real-world slide digitization workflows and predict system performance under real-world operational constraints.

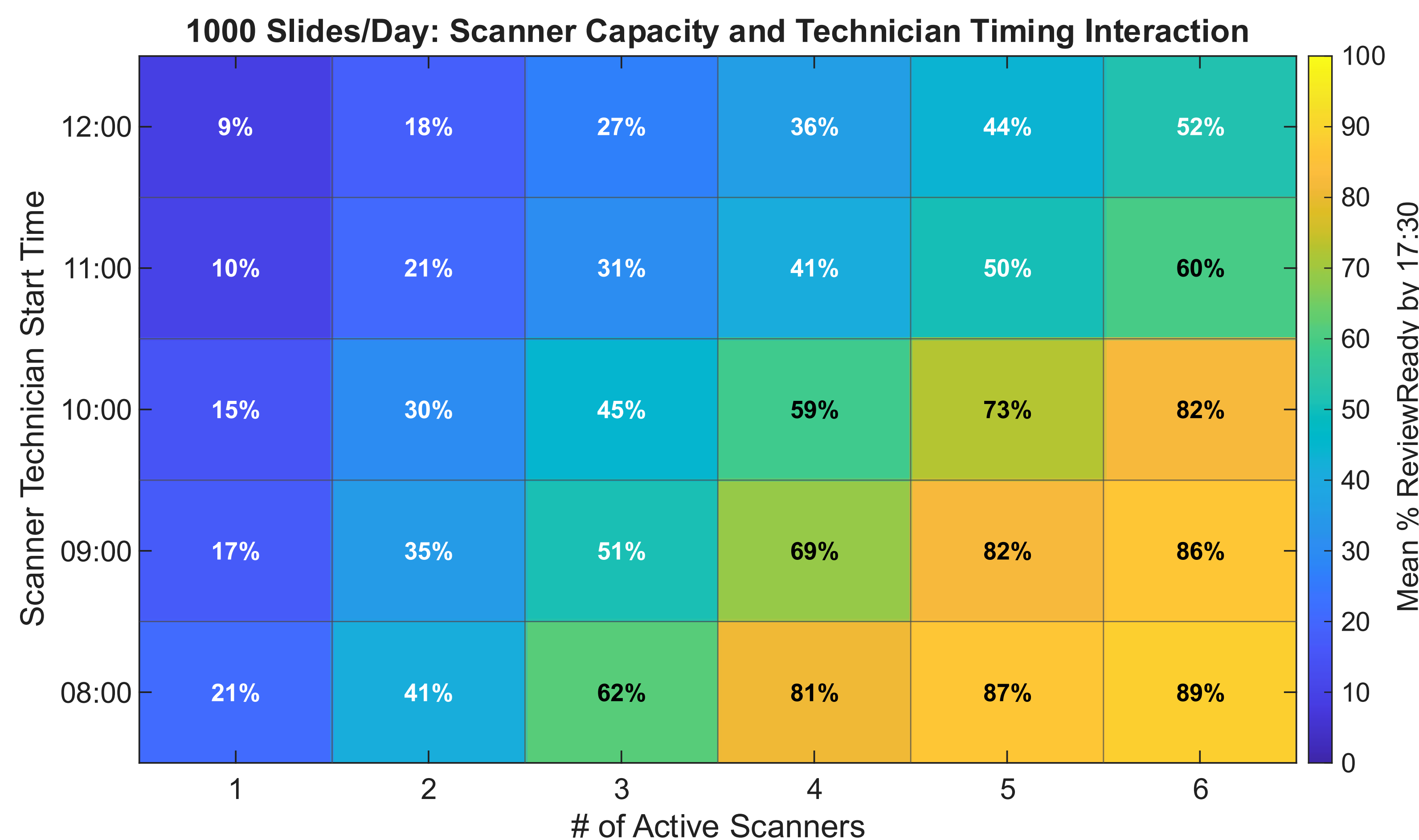
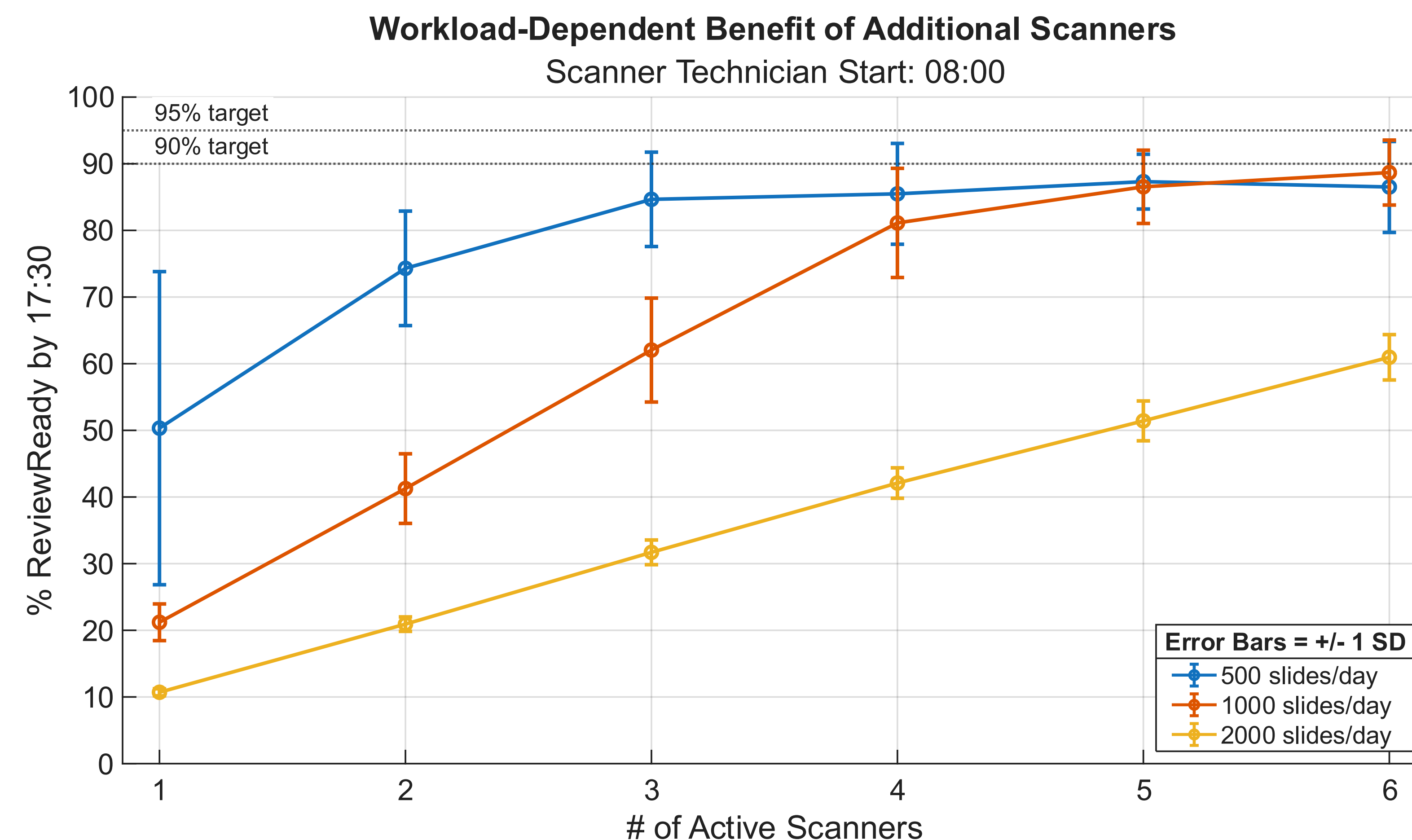
We implemented a DES model in **MATLAB Simulink** to simulate slide flow from glass-slide preparation through scanning, post-processing, and availability in the electronic medical record. Model inputs were derived from empirical workflow data, including **hourly slide arrival rates, slides per case, and scanner performance characteristics**. System constraints included **scanner technician availability windows and scanning throughput**. The framework enables parameterization of **staffing levels, scanner counts, and workload patterns** for scenario-based analysis.

METHODS



- Workloads:** 500, 1000, and 2000 target mean slides/day; stochastic daily realizations.
- Scanning:** 20-slide racks; lognormal scan time, median 58 s / mean 110 s per slide.
- Post-scan:** lognormal latency, median 20 min / P95 120 min.
- Pilot design:** 3 replications/scenario; matched workload realizations across resource configurations; endpoint = % ReviewReady by 17:30.

RESULTS

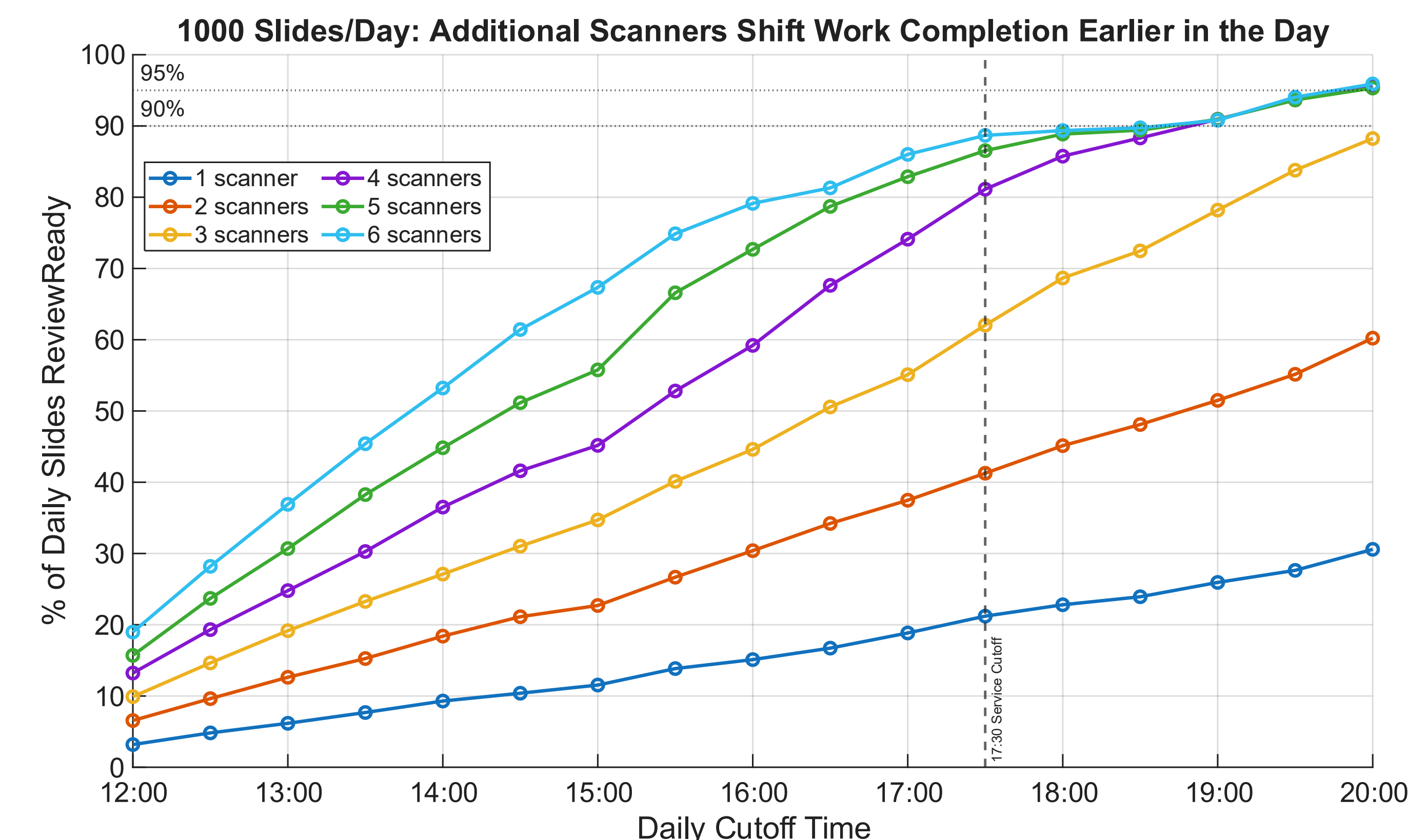


Why the plateau? Bottleneck migration: Once scanner capacity exceeds incoming scan-ready work, added scanners cannot overcome slide availability, transport/loading, or post-scan delay.

CONCLUSIONS

Take-home: Match equipment and labor to when slides arrive - not just total slides/day.

- Diminishing returns emerged at medium workload.** At ~1000 slides/day, ReviewReady by 17:30 rose 21.2% -> 41.3% -> 62.0% -> 81.1% with 1-4 scanners, then only to 86.5% and 88.7% with 5-6 scanners.
- Staffing timing changed the value of the same fleet.** With 6 scanners, an 08:00 scanner technician start achieved 88.7% ReviewReady by 17:30 versus 52.3% with a 12:00 start.
- Workload regime matters.** Low volume approached a plateau within the tested fleet, whereas high volume remained scanner-constrained through 6 scanners.



CONTACT / ACKNOWLEDGEMENTS / RESOURCES

- Future production analysis:** 30 replications/scenario with 95% confidence intervals.
- Future version features:** Cutoff-state bottleneck attribution, multiday/tiered turnaround policies, and protocol/sample aware stochastic scanning times.

GITHUB
PROJECT
REPOSITORY



INTERACTIVE
SIMULATION
MODEL